

Bias

- What is a bias in a neural network?

-> Bias is an additional parameter added to the weighted sum of inputs before applying the activation function.

For a neuron:

$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

where:

- x_i = input
- w_i = weight
- b = **bias**
- z = input to the activation function

Bias gives the neuron an extra value that it can learn independently of the inputs.

- Why do we add bias to a perceptron?

-> to **shift the decision boundary** .

Without bias:

$$z = w_1x_1 + w_2x_2$$

For a perceptron, the decision boundary is:

$$w_1x_1 + w_2x_2 = 0$$

This boundary is forced to pass through the **origin (0, 0)**.

When we add bias:

$$w_1x_1 + w_2x_2 + b = 0$$

Now the boundary can **move away from the origin**.

Weights control the slope/orientation of the decision boundary, while bias controls its position.

Weights

- What are weights in a perceptron?

-> **Weights** are learnable parameters that determine how important each input is to a perceptron.

- How do weights affect the output of a perceptron?

Each input is multiplied by its corresponding weight:

$$z = w_1x_1 + w_2x_2 + b$$

Then the result is passed through an activation function.

- **Large positive weight** → input has a strong positive influence.
- **Large negative weight** → input has a strong negative influence.
- **Weight close to 0** → input has little influence.

->

Perceptron Output

- A perceptron has inputs $x_1=2$, $x_2=3$, weights $w_1=0.5$, $w_2=1$, and bias $b=1$. Calculate the weighted sum (before activation).

Ans 5

Step function :

What is the step function and how does it work?

-> A **step function** is an activation function that converts the neuron's weighted sum into a **binary output**, usually **0 or 1**.

For a perceptron:

$$z = w_1x_1 + w_2x_2 + b$$

The step function is commonly defined as:

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

How does it work?

It checks whether the weighted sum crosses a **threshold**.

- If $z \geq 0 \rightarrow$ output 1
- If $z < 0 \rightarrow$ output 0

Activation Function

- Why do we need an activation function in a perceptron?
-> perceptron calculates Z. Activation function brings it in a range and shows as a output of the neuron.

Why is it needed?

The main reasons are:

1. It determines whether the neuron should be activated or not.
2. It introduces non-linearity in neural networks, allowing them to learn complex patterns.
3. In a perceptron, it can convert the weighted sum into a suitable output such as 0 or 1 for binary classification.

For example, with a step function:

$$f(z) = \begin{cases} 1 & z \geq 0 \\ 0 & z < 0 \end{cases}$$

- So, the activation function basically decides what the neuron outputs.

- Name three common activation functions.

Sigmoid

ReLU (Rectified Linear Unit)

Tanh (Hyperbolic Tangent)